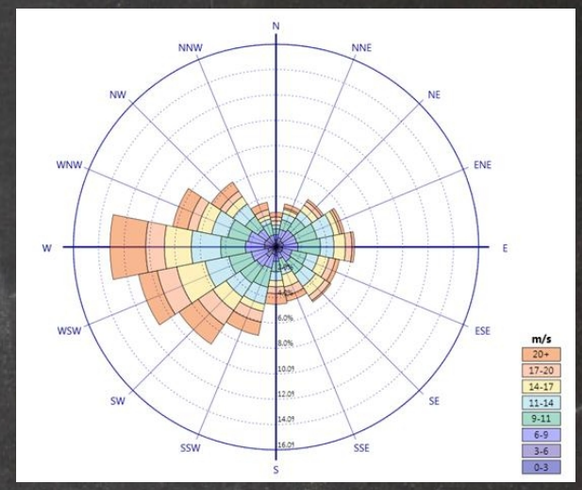
Wind Farm Layout Optimization Competition

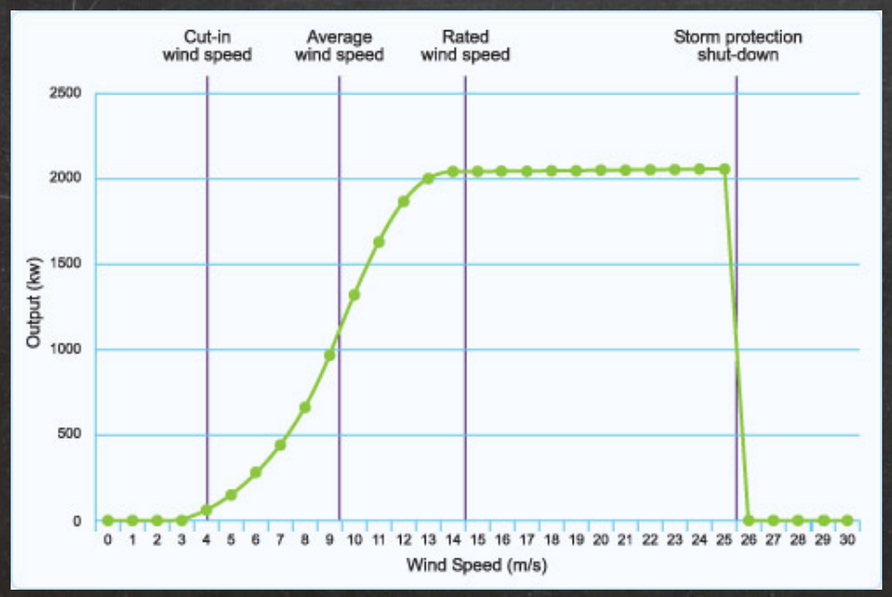
2nd Edition

Organizers: D. Wilson, S. Cussat-Blanc, S. Rodriguez, K. Veeramachaneni

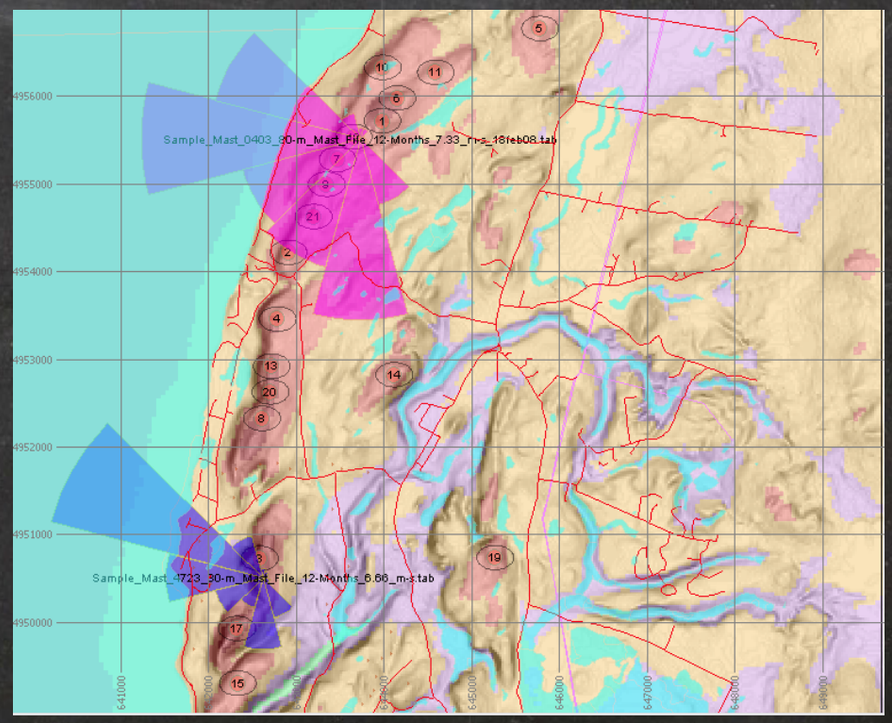
Wind farm layout optimization

- Objectives:
  - Optimize a layout of hundreds of wind turbines
  - Optimizer in the middle of a design loop that include humans
  - Multiple local and global design considerations:
    - Wind distribution and wake effects
    - Safety constraints
    - Turbine power curve
    - Economic constraints (construction and maintenance cost, cable and road optimization, etc.)
    - Topographic constraints (lakes, mountains, roads, buildings, etc.)
    - “Human” constraints (noise, visual, etc.)





Source: pfr



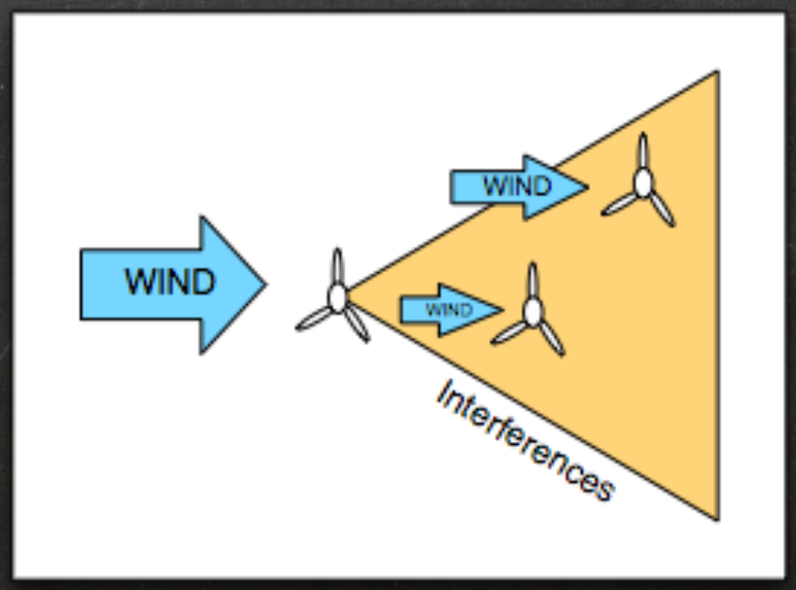
Source: AWS OpenWind

Wind farm layout optimization

- Energy capture model from [Kusiak & Song 2010]:
  - Each turbine generates interferences for other turbines behind it
  - Based on a Weibull distribution, reduced by the wake generated per the turbines:

$$E^i[\eta] = \int_{\theta} P(\theta) \int_{v} p^{\theta}_{v}(v, c_i, k_i, x_i, y_i, X, Y)\beta^i(v)$$

    - *P(θ)*: Wind flow probability from direction θ
    - *pθv(v, ci, ki, xi, yi, X, Y)*: Weibull distribution with wake loss reduction
    - *β(v)*: turbine power curve
  - Provides the global energy capture of the layout and for each turbines



Wind farm layout optimization

- Energy capture model from [Kusiak & Song 2010]:
  - Implemented in the open-source WindFLO API
    - Matlab
    - C++
    - Java
    - Python
  - Available online: https://github.com/d9w/WindFLO
  - Provides a set of random and real test scenarios



Competition rules

- Objective: find the best layout that minimize the cost of energy

$$fitness = \frac{\overbrace{\left(c_t * n + c_s * floor(\frac{n}{m})\right)}^{\text{Construction cost}} \overbrace{\left(\frac{2}{3} + \frac{1}{3} * e^{-0.00174n^2}\right)}^{\text{Economies of scale}} + \overbrace{c_{OM} * n}^{\text{Yearly operating costs}}}{\left.\left(1-(1+r)^{-y}\right)\middle/ r\right\}\ \text{Interests}} * \underbrace{\frac{1}{8760 * P}}_{\substack{\text{Yearly} \\ \text{power} \\ \text{output}}} + \overbrace{\frac{0.1}{n}}^{\substack{\text{Farm size} \\ \text{coefficient}}}$$

$c_t = 750{,}000$ turbine cost (usd)

$c_s = 8{,}000{,}000$ substation cost (usd)

$m = 30$ turbines per substation

$r = 0.03$ Interest rate

$y = 20$ Farm lifetime (years)

$c_{OM} = 20{,}000$

n Number of turbines

P Farm energy output

Competition rules

- Constraints:
  - 5 unknown layouts to optimize
  - Obstacles
  - Optimize both the number of turbines and their positions
  - 10,000-evaluations total limit
- Ranking system:
  - Competitors ranked on each scenario independently
    - Best: 10 pts
    - 2nd: 6 pts
    - 3rd: 4 pts
    - 4th: 3 pts
    - 5th: 2 pts
    - 6th: 1 pt
    - 7+: 0 pt
  - Competitors final ranking by summing the points

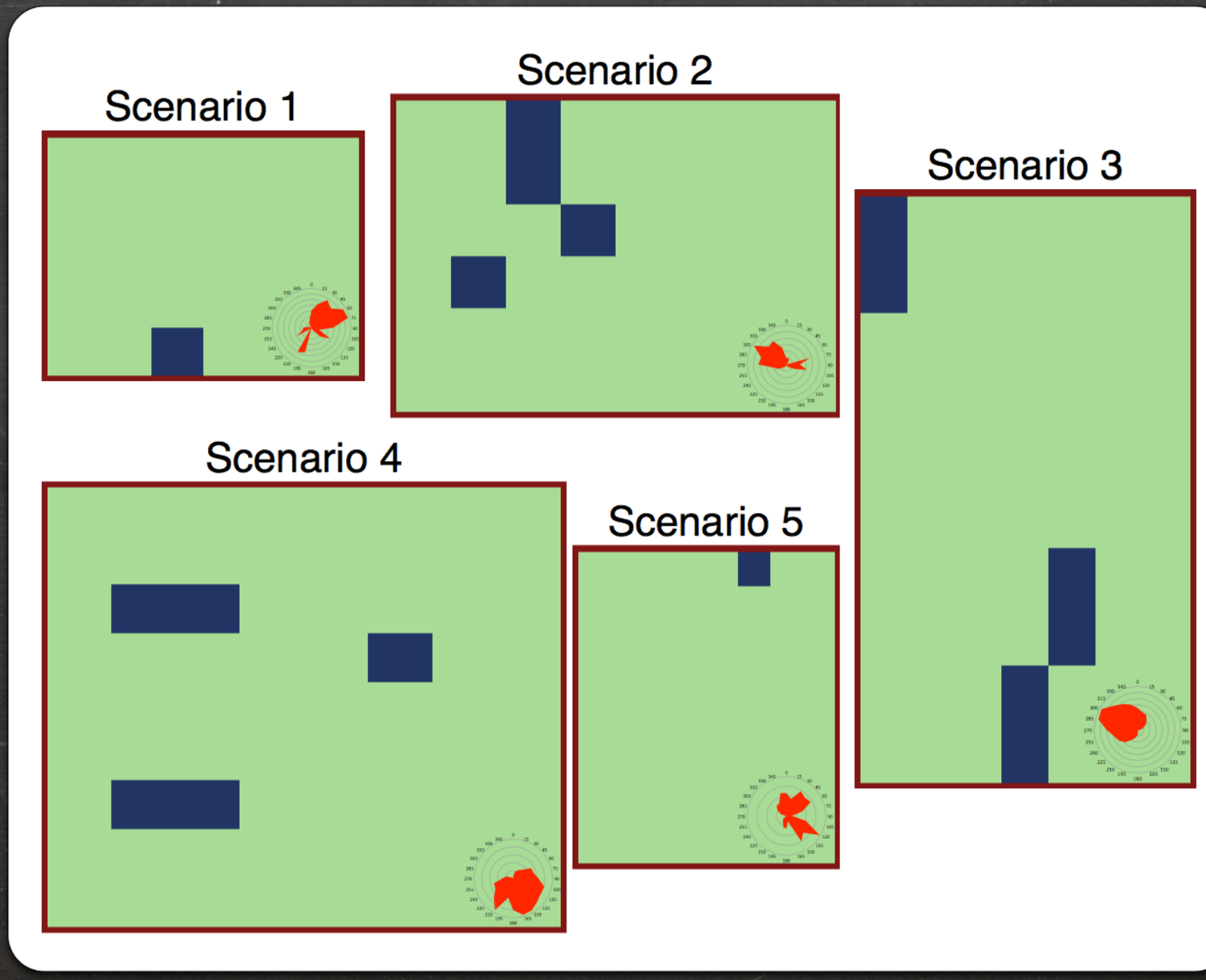
Competition schedule

- 3 stages:
  - Offline evaluations
    - Wake model
    - Cost of energy function
    - Use example (simple GA)
    - Set of scenarios
  - Online competition on testing layouts
    - Unlimited evaluations
    - In-the-cloud evaluations
    - Unknown scenarios
    - Online leaderboard
  - Online competition
    - 10,000-limit evaluations
    - In-the-cloud evaluations
    - 5 new unknown scenarios
    - 3 days to run the algorithm
    - No leaderboard (Suspense!)



Layouts

- The 5 final layouts:

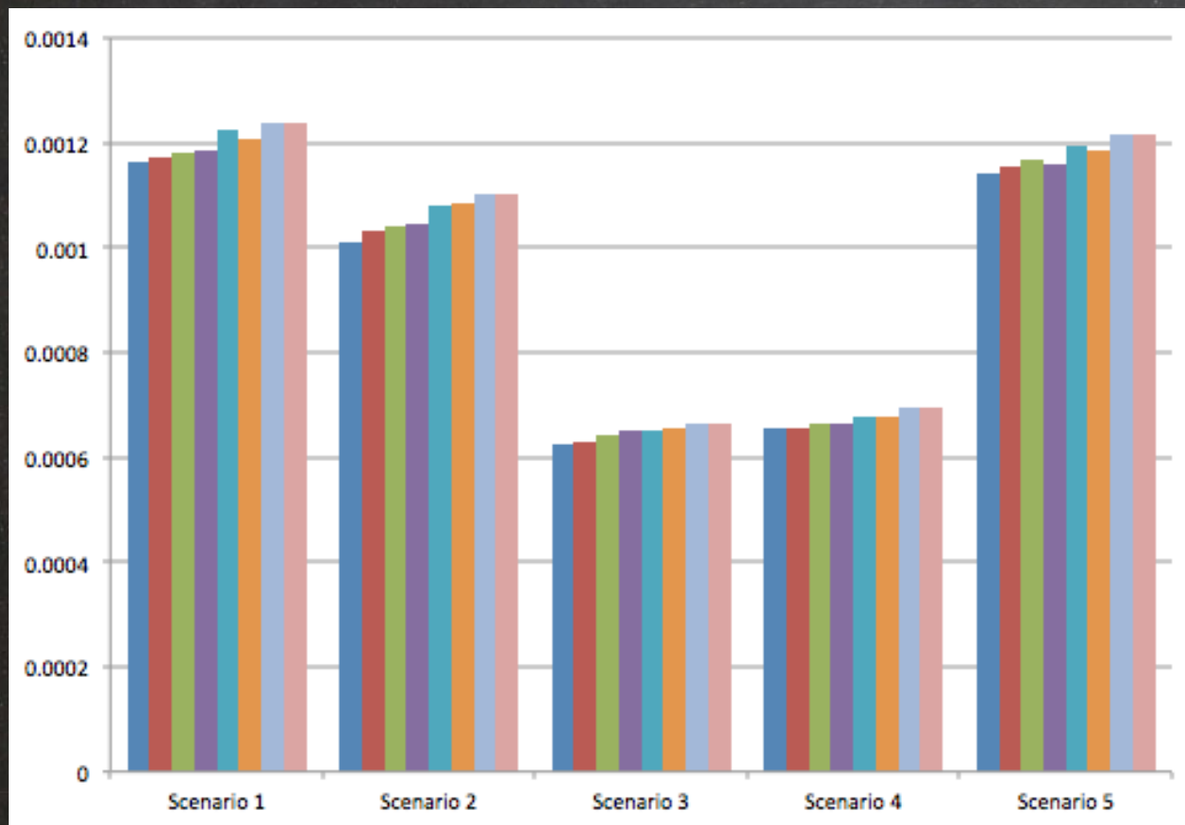


Results

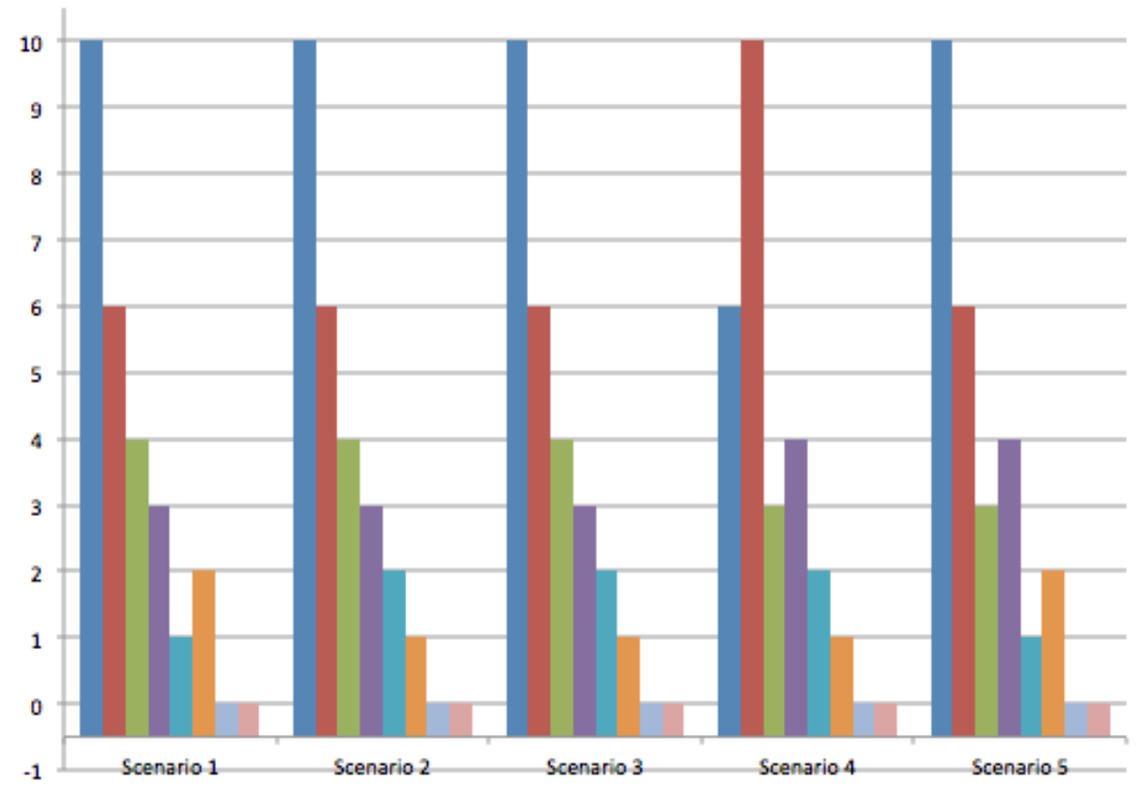
- This year, 8 competitors (+400%!!!)
  - Wonderfull help from them to improve the API
  - THANK YOU FOR YOUR HELP

Results

- This year, 8 competitors (+400%!!!)
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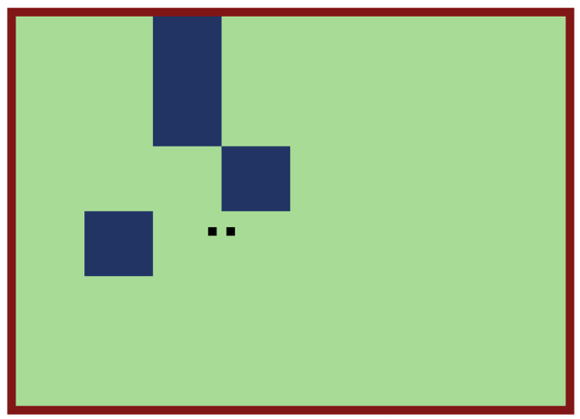


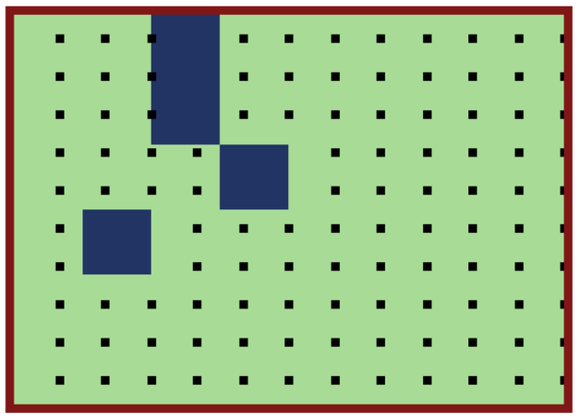
Cost of energy
(lower is better)

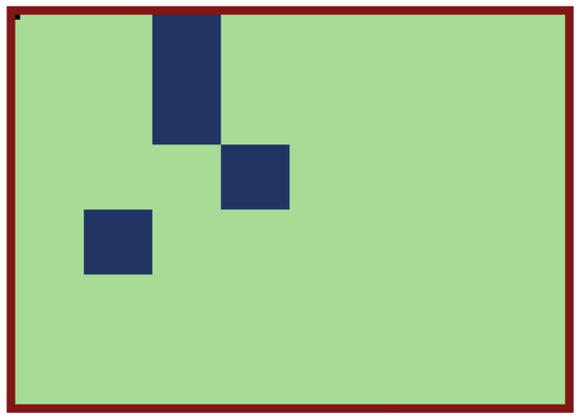


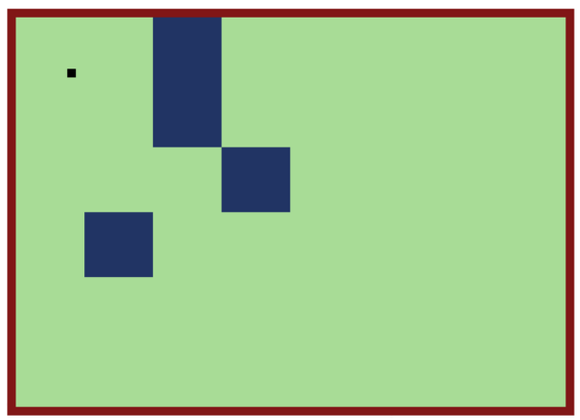
Points
(higher is better)

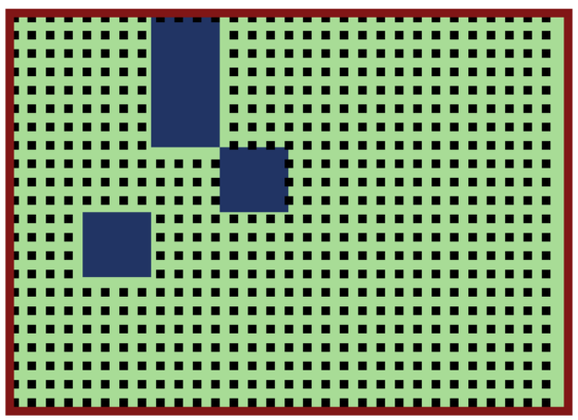
Example of solutions

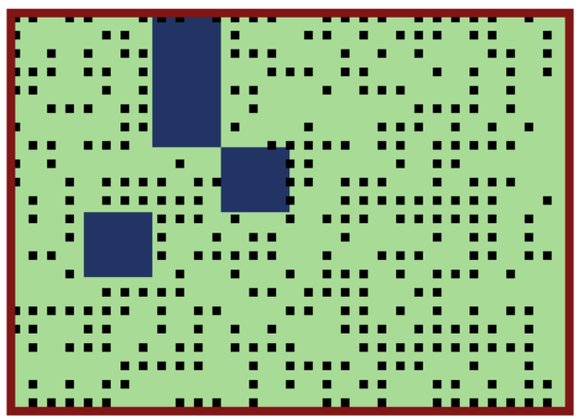












Results

- Final ranking

	Scenario 1	Scenario 2	Scenario 3	Scenario 4	Scenario 5	Score
#1	10	10	10	6	10	46
#2	6	6	6	10	6	34
#3	4	4	4	3	3	18
#4	3	3	3	4	4	17
#5	1	2	2	2	1	8
#6	2	1	1	1	2	7
#7	0	0	0	0	0	0
#8	0	0	0	0	0	0

Results

- Final ranking

	Scenario 1	Scenario 2	Scenario 3	Scenario 4	Scenario 5	Score
#1	10	10	10	6	10	46
#2	6	6	6	10	6	34
#3	4	4	4	3	3	18
#4	Brian Goldman					17
#5	Michael Mayo					8
#6	Krzysztof Michalak					7
#7	Sergio Rivera					0
#8	Fabricio Loor					0

Results

- Final ranking

	Scenario 1	Scenario 2	Scenario 3	Scenario 4	Scenario 5	Score
#1	10	10	10	6	10	46
#2	6	6	6	10	6	34
#3	Ahmed Kheiri					18
#4	Brian Goldman					17
#5	Michael Mayo					8
#6	Krzysztof Michalak					7
#7	Sergio Rivera					0
#8	Fabricio Loor					0

Results

- Final ranking

	Scenario 1	Scenario 2	Scenario 3	Scenario 4	Scenario 5	Score
#1	10	10	10	6	10	46
#2	Ilya Loshchilov & Frank Hutter					34
#3	Ahmed Kheiri					18
#4	Brian Goldman					17
#5	Michael Mayo					8
#6	Krzysztof Michalak					7
#7	Sergio Rivera					0
#8	Fabricio Loor					0

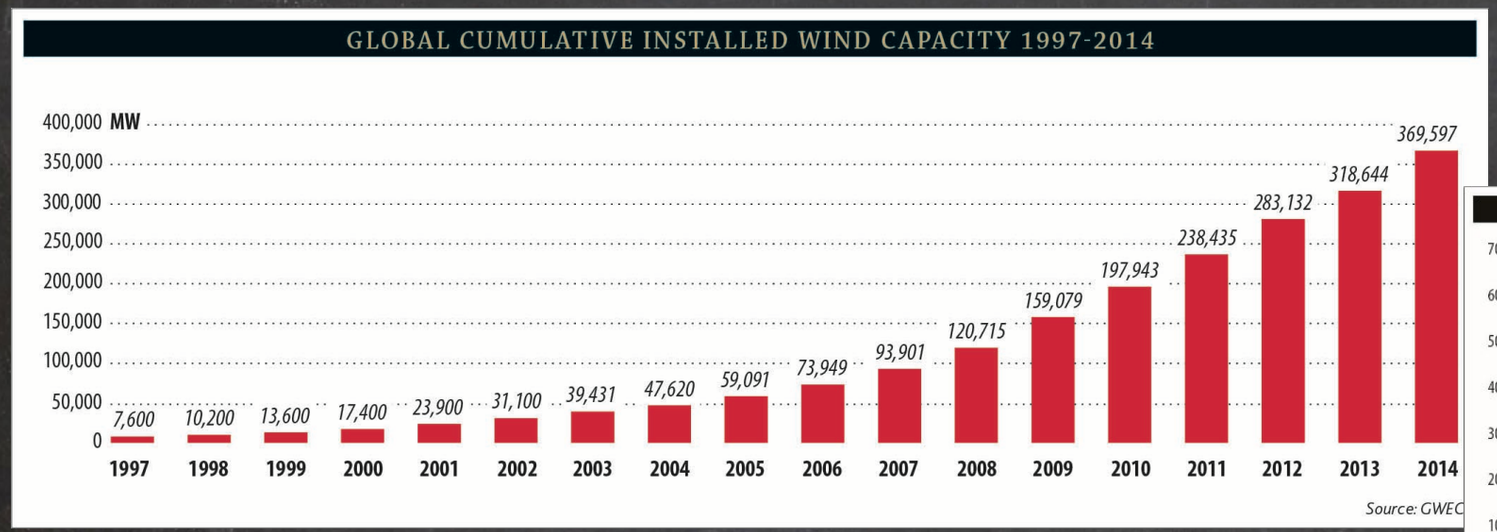
Results

- Final ranking

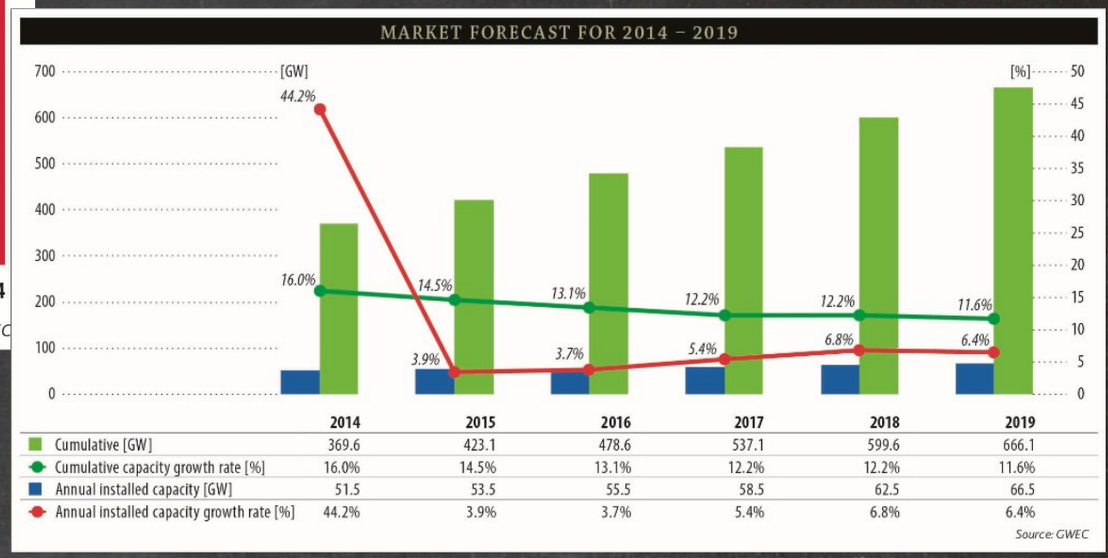
	Scenario 1	Scenario 2	Scenario 3	Scenario 4	Scenario 5	Score
#1	Carlos Segura - CIMAT Team					46
#2	Ilya Loshchilov & Frank Hutter					34
#3	Ahmed Kheiri					18
#4	Brian Goldman					17
#5	Michael Mayo					8
#6	Krzysztof Michalak					7
#7	Sergio Rivera					0
#8	Fabricio Loor					0

What's next?

- Wind energy is a growing market:



Installed wind capacity 1997-2014



Forecast 2014-2019

- Next year, it won't be that easy!
  - Different turbine types?
  - Cable and road networks?
  - Terrain elevation?

